

A novel intrusion detection framework for industrial IoT: GCN-GRU architecture optimized with ant colony optimization

Mahdi Mir^{a,*}, Mohammad Trik^{b,*}

^a Department of Electrical Engineering, Ferdowsi University of Mashhad, Mashhad, Iran

^b Department of Computer Engineering, Boukan Branch, Islamic Azad University, Boukan, Iran

ARTICLE INFO

Keywords:

GCN-GRU Architecture
Industrial IoT
Deep learning optimization
Intrusion detection
Ant colony optimization

ABSTRACT

The swift proliferation of IIoT ecosystems has highlighted the essential requirement for effective Intrusion Detection System (IDS) to protect crucial infrastructures. This research presents a novel hybrid IDS that combines Graph Convolutional Networks (GCN) with Gated Recurrent Units (GRU), optimized by the Ant Colony Optimization (ACO) method, a bio-inspired meta-heuristic based on ant foraging behavior. This method automates hyperparameter adjustment, overcoming the constraints of conventional human optimization techniques. The proposed system utilizes GCN for structural feature extraction and GRU for sequential pattern analysis, facilitating thorough anomaly detection in IIoT traffic. The ACO-optimized IDS surpasses traditional optimization methods, including Genetic Algorithms, by attaining quicker convergence and enhanced performance metrics. Notwithstanding its effectiveness, the computational burden of the optimization approach necessitates additional enhancement. Experimental assessments of the EDGE-IIOTSET, CICAPT-IIoT, and WUSTL-IIoT datasets reveal detection accuracies of 97 % for the majority of attack scenarios, alongside improved scalability and diminished processing requirements. This study emphasizes the capability of integrating sophisticated neural architectures with nature-inspired optimization to enhance Industrial Internet of Things (IIoT) security against advancing cyber threats.

1. Introduction

By integrating IoT technologies with industrial control systems (ICS), the IIoT has revolutionized industrial operations since its inception in the early 2000s, facilitating automation, efficiency, and connectivity [1]. Initially focused on consumer applications, IIoT migrated to industrial sectors by the mid-2010s and was driven by advances in sensor technologies, cloud computing, and machine-to-machine communication. The high volume and speed of IIoT traffic require low-latency and scalable solutions, as demonstrated by the Service Allocation Framework in IoT Networks and inspired the optimized IDS in this work [2,3]. Today, the IIoT supports critical infrastructure, including power grids, industrial plants, and transportation systems, enabling real-time monitoring, predictive maintenance, and data-driven decision-making [4].

Given the contribution of IIoT networks to critical services and economic stability, the importance of protecting them is paramount. Unlike commercial networks, industrial systems emphasize operational stability, as even small disruptions can lead to significant consequences, including production downtime, environmental hazards, or loss of life. Historical cyberattacks, including the Stuxnet

* Corresponding authors.

E-mail addresses: Mahdimir.ir@gmail.com (M. Mir), m.trik@iau-tnb.ac.ir (M. Trik).

worm in 2010 and the Triton malware in 2017, have demonstrated the catastrophic potential of attacks on industrial control systems and underscore the need for specialized security protocols [5]. Projections indicate that unpatched cybersecurity vulnerabilities in IIoT could result in global economic losses of more than \$90 trillion by 2030, highlighting the need for comprehensive protection measures. IDS are essential for IIoT security [6], including proactive monitoring and threat identification to maintain the confidentiality, integrity, and availability (CIA) of network operations [7]. Unlike conventional firewalls, IDS examine network traffic to detect anomalies and malicious activity, making them essential for mitigating sophisticated attacks that exploit the distinctive characteristics of the IIoT, including resource-constrained devices and diverse protocols [8].

Building efficient intrusion detection systems for the IIoT context presents several obstacles. The diversity of IIoT devices, such as sensors and actuators, generates diverse data formats and communication protocols, thus hindering the development of universal detection models. The significant volume and velocity of IIoT traffic requires scalable and computationally efficient solutions. Improving deep learning models, which have great potential in IDS applications, poses another challenge, as manually tuning the above parameters such as learning rate, layer configuration, and tuning parameters is labor-intensive and often suboptimal [9]. These issues require the development of unique IDS frameworks that integrate powerful neural architectures with automated optimization approaches to improve detection accuracy and efficiency. Fig. 1 shows a typical IIoT architecture that emphasizes cloud infrastructure, edge nodes, and device layers, where sensors, actuators, and communication protocols enable data transmission [10].

This research is motivated by the need to overcome these limitations while leveraging deep learning and bio-inspired optimization. Deep learning has revolutionized intrusion detection by enabling automated feature extraction and robust pattern recognition in complex network environments. However, its computational complexity and hyperparameter optimization requirements often prevent practical deployment in resource-constrained IIoT settings. Hyperheuristic optimization techniques, inspired by natural processes, have been effective in automating hyperparameter tuning, reducing human effort, and improving model performance. By integrating these paradigms, this study develops a scalable and efficient IDS that is optimized with ACO for fast convergence and high detection accuracy by using GCN to model structural relationships in network traffic and GRU to capture temporal dependencies. The novelty of this work lies in the integration of the GCN-GRU architecture with ACO, a combination that has been less explored in IIoT intrusion detection. Unlike conventional convolutional neural networks (CNNs), GCNs represent IIoT traffic in a graph format and effectively capture device interactions and structural anomalies that are critical for detecting advanced attacks. GRU provides a simple alternative to long-term short-term memory (LSTM) networks that reduces processing demands while maintaining strong temporal analysis capabilities. By mimicking the search behavior of ants, ACO balances exploration and exploitation, outperforming traditional methods such as particle swarm optimization (PSO) in terms of convergence speed and solution quality. This hybrid approach addresses the computational limitations of IIoT while increasing detection effectiveness, making it suitable for real-time applications. Key achievements of this study include:

- Accurate analysis of IIoT traffic characteristics, highlighting industrial network differences, and tailored security strategies.

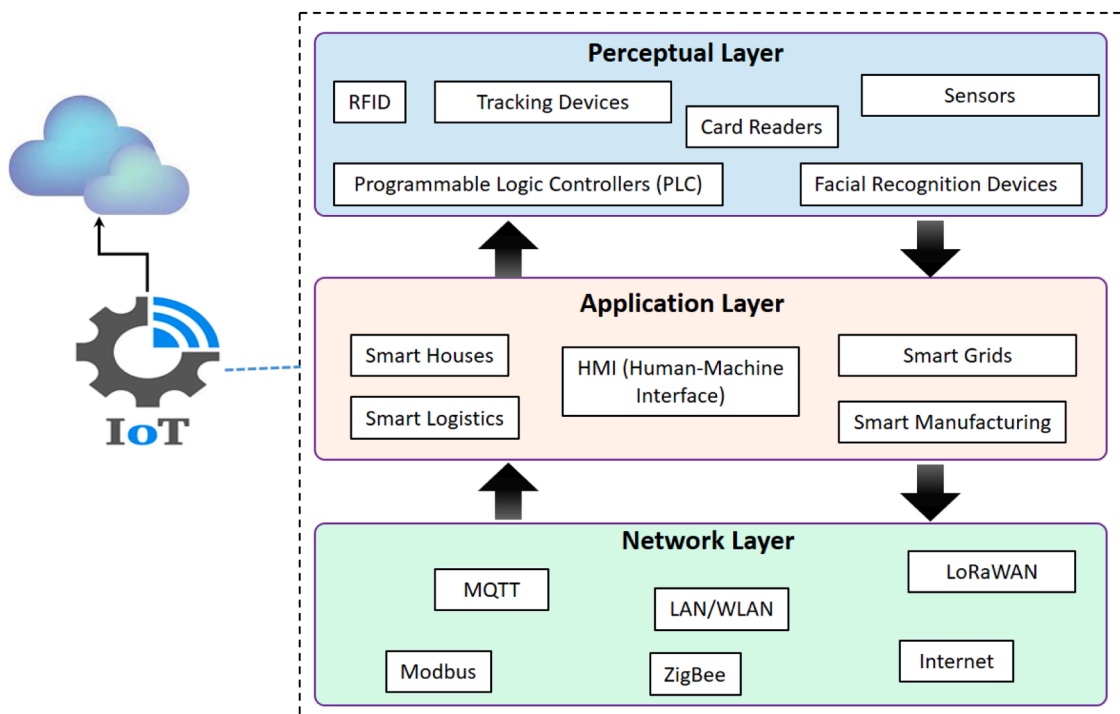


Fig. 1. A typical architecture for the IIoT.

- A novel IDS using GCN-GRU for structural and temporal analysis, achieving high detection accuracy across attack types.
- ACO-based hyperparametric tuning, reducing training time and addressing the limitations of manual tuning.

This paper is organized as follows: [Section 2](#) reviews the related literature, [Section 3](#) describes the proposed GCN-GRU and ACO method in detail, [Section 4](#) presents the experimental setup and results, [Section 5](#) compares the performance, and [Section 6](#) concludes with future directions.

2. Related work

The IIoT has swiftly proliferated, improving daily life, industry, and society through the interconnection of digital devices and physical systems via sensors and actuators. Despite producing extensive data, resource-constrained devices encounter problems stemming from heterogeneity, jeopardizing continuous operations. This requires the implementation of comprehensive security solutions, resulting in sophisticated IDS that utilize machine learning and deep learning techniques. Recent studies, such as [\[11\]](#), examine these methodologies, classifying them into agent insertion, detection strategies, and security issues to effectively mitigate IIoT risks.

The ubiquity of security vulnerabilities in Internet of Things (IoT) applications is a significant threat to enterprise systems, necessitating advanced and dependable protection solutions to address emerging and developing threats. The research paper [\[12\]](#) introduces a resilient deep learning model named AttackNet for the detection and classification of diverse botnet attacks in IIoT, utilizing a matching-based CNN-GRU architecture.

The research [\[13\]](#) develops an intrusion detection model employing feature engineering and machine learning for the security of IIoT. They integrate the isolated forest (IF) with the Pearson correlation coefficient (PCC) to diminish computing expense and prediction duration. They utilize Principal Component Analysis to identify the most suitable attributes. PCC and IF are utilized interchangeably (PCCIF and IFPCC). A random forest (RF) classifier is employed to improve the efficacy of the IDS.

Researchers at [\[14\]](#) have introduced a federated learning (FL) methodology for identifying undesirable intrusions to safeguard IoT networks. This method guarantees privacy and security by aggregating data from local IoT devices. Local IoT clients only transmit parameter updates to a central global server, which consolidates them and disseminates an enhanced detection algorithm. Subsequent to each iteration of machine learning training, each IoT client obtains an updated model from the global server and trains its local dataset, allowing IoT devices to preserve their privacy while enhancing the overall model.

In [\[15\]](#), researchers introduced an advanced detection technique for identifying cyberattacks in industrial IoT networks. Their proposed model employs the singular value decomposition (SVD) method to diminish data characteristics and enhance detection outcomes. They employed the Synthetic Minority Over-sampling Technique (SMOTE) to mitigate overfitting and underfitting issues that result in biased classification. A variety of machine learning and deep learning methods are employed to classify data for both binary and multiclass classification.

In [\[16\]](#), three distinct models were introduced for intrusion detection in IIoT networks utilizing deep learning architectures: CNN, LSTM, and a hybrid model combining CNN and LSTM. A study utilizing the UNSW-NB15 and X-IIoTID datasets identified and compared normal and aberrant data with existing literature after binary and multi-class classification. The CNN + LSTM hybrid model attained the highest accuracy for intrusion detection across both datasets among the recommended models.

In [\[17\]](#), researchers introduce a multi-stage deep learning (MSDL) approach aimed at improving intrusion detection in IIoT networks by tackling the class imbalance issue. They assess the efficacy of their methodology utilizing two significantly imbalanced datasets: X-IIoTID and WUSTL-IIoT.

Notwithstanding advancements in IDS, there is an urgent requirement for highly precise, adaptable, and scalable solutions tailored for cloud-based IIoT contexts. Driven by this demand, researchers at [\[18\]](#) have introduced a sophisticated AI-based IDS that utilizes LSTM networks and GRU. [Table 1](#) offers a succinct comparison of current IIoT intrusion detection techniques, emphasizing both their benefits and drawbacks.

3. Proposed method

Deep neural networks are vital for developing IDS tailored for IIoT environments. A key challenge, as noted in [Section 2](#), is optimizing their hyperparameters. This study introduces an innovative IDS framework integrating GCN with GRU, enhanced by ACO, a meta-heuristic inspired by ant foraging behavior. The GCN-GRU architecture is designed, trained, and evaluated, but its performance

Table 1
Assessment and comparison of earlier works.

Ref.	Proposed Method	Advantages	Disadvantages
[12]	AttackNet (CNN-GRU)	High accuracy (99.75 %) for botnet detection	Limited focus on emerging IIoT threats
[13]	IF-PCC with Random Forest	Reduces computational cost using feature selection	Lacks real-time adaptability
[14]	Federated Learning (FL) with IoT Devices	Preserves privacy via local updates	High computational overhead
[15]	SVD-SMOTE with ML/DL Models	Mitigates overfitting with balanced data	Requires evaluation on newer datasets
[16]	CNN-LSTM Hybrid Model	High accuracy on UNSW-NB15 and X-IIoTID datasets	Limited scalability for large-scale IIoT
[17]	Multi-Stage Deep Learning (MSDL)	Addresses class imbalance in IIoT datasets	High computational complexity
[18]	LSTM-GRU for Cloud-Based IIoT	Effective for cloud environments	Manual hyperparameter tuning

depends on fine-tuning hyperparameters like learning rate, neuron count, and dropout rates. Manual tuning via trial-and-error is time-consuming and often yields suboptimal results.

To overcome this, ACO automates hyperparameter optimization for the GCN-GRU model. Known for balancing exploration and exploitation, ACO efficiently navigates complex, high-dimensional search spaces, avoiding local optima and achieving rapid convergence, outperforming other meta-heuristics in deep learning tasks. The GCN extracts structural features from IIoT traffic data, capturing device interactions, while GRU processes sequential data to detect temporal anomalies and attack patterns. Together, they enhance detection accuracy and computational efficiency. Fig. 2 illustrates the IDS training and evaluation phases. During training, ACO optimizes the GCN-GRU model, automating hyperparameter tuning, reducing computational demands, and improving efficiency compared to traditional manual methods.

3.1. The role of ACO in the proposed method

The ACO method was selected for its ability to mimic ants' intelligent foraging, enabling efficient navigation in complex search spaces. This meta-heuristic balances global exploration and focused exploitation, making it ideal for optimizing deep learning models. During exploration, ACO employs probabilistic algorithms to scan the search space, while in exploitation, pheromone-guided refinement accelerates convergence. The proposed method uses ACO to automatically tune hyperparameters of the GCN-GRU architecture, including learning rate, neuron counts, dropout rates, and pooling dimensions, enhancing IDS efficiency and reducing manual optimization's computational burden. The GACO framework, shown in Fig. 2, includes three components: a preprocessor for noise filtering and feature extraction, and a GCN-GRU analyzer engine that detects spatial and temporal anomalies in IIoT traffic.

Controller and Manager: This component oversees the optimization process, utilizing ACO to determine the ideal hyperparameters for the GCN-GRU model [19]. The controller optimizes the training phase, allowing the model to attain maximum performance with lowest resource utilization. The amalgamation of ACO with the GCN-GRU architecture yields numerous advantages:

Automated Hyperparameter Optimization: ACO eliminates the necessity for human modifications, reducing trial-and-error endeavors and enhancing productivity.

The enhanced GCN-GRU model attains heightened precision in detecting intrusions, anomalies, and attacks.

Accelerated Training Process: ACO's effective convergence diminishes the duration necessary for training, enhancing the system's practicality for real-world applications.

The proposed IDS architecture utilizes the analytical prowess of GCN-GRU networks and the optimization skills of ACO to provide a reliable and efficient solution for safeguarding IIoT settings. This strategy addresses the limitations of traditional IDS techniques, markedly enhancing detection efficacy while minimizing processing requirements.

3.2. Preprocessor component

Fig. 3 illustrates the data pretreatment and optimization pipeline for the GACO classification system, designed to secure future IIoT networks. Using TON_IoT and NOTURIN-20S datasets, preprocessing starts with numerical encoding of textual data, followed by normalization to a uniform scale. Network features are extracted from traffic flows and converted into numerical formats for the GCN-GRU architecture. In the optimization phase, initial data features are randomly placed, and a novel "Cke noan" halting criterion is evaluated. If unmet, fitness is calculated using equations (0 or 25), updating the best position. The "tlal prosoever" method enhances depth space, followed by iterative position updates until the criterion is satisfied, enabling GACO classification.

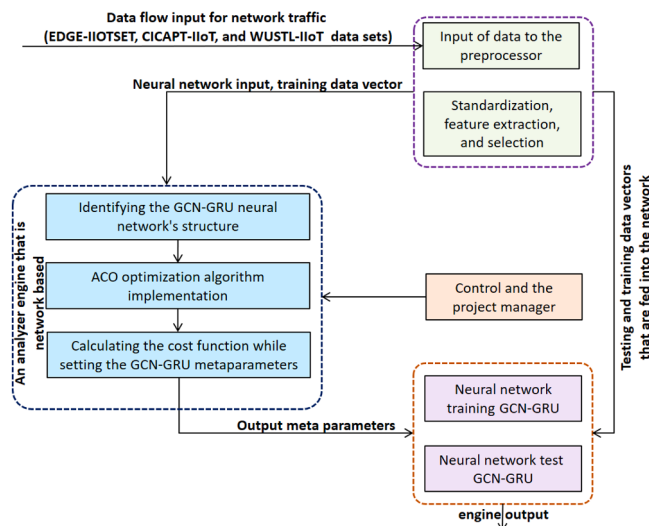


Fig. 2. GACO method architecture.

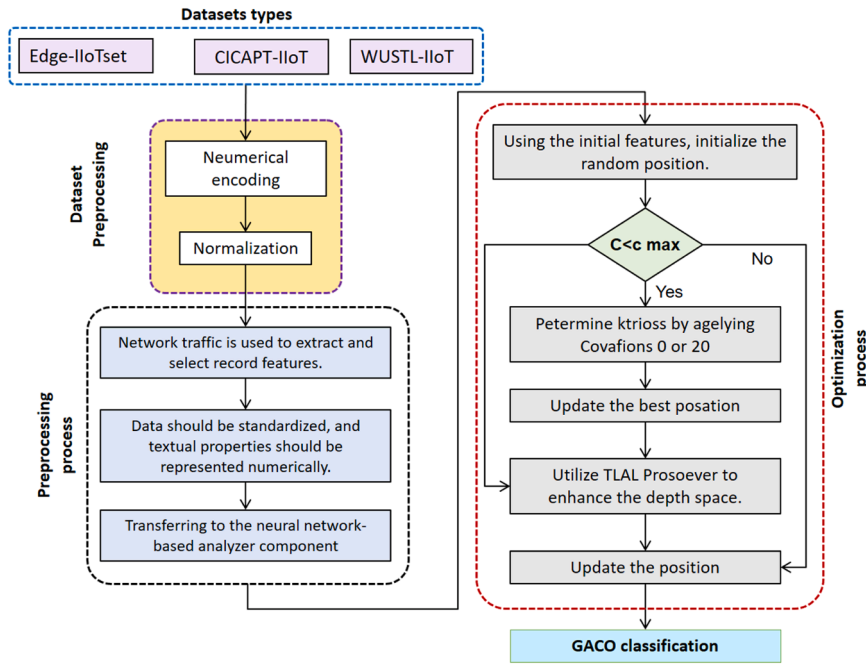


Fig. 3. Preprocessing process.

a) Conversion of Data to Numerical Format: This research incorporates input datasets that include temporal, binary, nominal, and numerical variables, illustrating the intricacy of future IIoT environments. Textual data is transformed into numerical features with advanced tag encoding techniques, facilitating smooth integration with the computational architecture of neural networks. The cost function of the GACO framework balances accuracy and resource efficiency, inspired by fair scheduling in IoT networks, which is critical for resource-constrained IIoT environments [19].

b) Standardization: A custom normalization procedure is implemented to provide uniformity across features in this prospective system. This study used a bounded standardization function, scaling data within a fixed range (e.g., [0, 1]), departing from traditional Z-score procedures. The equation is reformulated as:

$$Z_i = \frac{X_i - \min(X)}{\max(X) - \min(X)} \quad (1)$$

where X_i represents the input value, and $\min(X)$ and $\max(X)$ signify the minimum and maximum values of the dataset, respectively. This method expedites neural network convergence and mitigates the influence of outliers, essential for the critical IIoT of the future. Recent advances, such as the HIDIM framework, resolve class imbalances and hierarchical dependencies in IIoT intrusion detection and complement the GCN-GRU approach for robust detection [20].

c) Selection of Features: reduces dataset dimensionality while preserving predictive power in preprocessing. Principal Component Analysis (PCA) identifies key features as eigenvectors of the covariance matrix with maximum variance, retaining significant components. Simulations using Modbus protocols identified 23 critical features for intrusion detection, with PCA refining the EDGE-IIOTSET dataset to 30 features, boosting efficiency and accuracy [21].

3.3. Neural network analyzer using GCN-GRU architecture

The neural network analyzer serves as the foundation of the GACO framework, an advanced technology designed to safeguard the IIoT networks of the future. This sophisticated component is responsible for analyzing intricate datasets, identifying trends, and categorizing network traffic as benign or harmful with exceptional accuracy. Utilizing GCN and GRU, the analyzer effectively captures spatial and temporal relationships in IIoT traffic, hence enabling robust intrusion detection in dynamic contexts. Federated learning enhances IIoT security, as demonstrated with the Byzantine Resilient Framework and supports the GCN-GRU distributed model for intrusion detection [22]. The GACO methodology consists of a carefully structured pipeline comprising three essential phases: hyperparameter optimization of the GCN-GRU architecture utilizing the ACO algorithm, development of a hybrid neural network model for efficient intrusion detection, and assessment of the final model's performance to satisfy the stringent requirements of

next-generation IIoT security.

3.3.1. Hyperparameter optimization of GCN-GRU using ACO

Hyperparameter optimization is critical for hybrid GCN-GRU architectures in deep learning, requiring precise tuning of layers, neurons, learning rates, and dropout rates to maximize intrusion detection performance in IIoT. Manual tuning is time-consuming and often yields suboptimal results for complex IIoT traffic. This study employs the ACO algorithm, a nature-inspired meta-heuristic, to efficiently navigate the hyperparameter search space. Non-intrusive load monitoring (NILM) can be used as a complementary method for anomaly detection in IIoT environments, as demonstrated by analyzing frequency signals for smart appliances [23]. Doubling neurons in the final two GCN layers enhances spatial pattern detection, while the GRU layer excels at capturing temporal dependencies, making the model adept at identifying time-dependent anomalies in IIoT traffic, as shown in Fig. 4.

ACO initializes a population of hyperparameter solutions, refined through exploration and exploitation phases. The GCN-GRU architecture excels in capturing spatiotemporal anomalies, similar to multi-head attention models for industrial sensor anomaly detection, which ensures accurate threat detection in IIoT [24]. Solution updates follow a mathematical model based on ant pheromone communication.

$$X_i^{t+1} = X_i^t + \eta \cdot (Ph_{best} - X_i^t) + \varsigma \cdot (Ph_{global} - X_i^t) \quad (2)$$

X_i^{t+1} denotes the revised position of the i th solution at iteration $t + 1$, X_i^t signifies the current position, Ph_{best} indicates the optimal position discovered by the i th ant, and Ph_{global} represents the global best position among all ants. The parameters η and ς govern the equilibrium between local and global search dynamics, facilitating an effective exploration of the hyperparameter space.

3.3.2. Cost function for ACO

The ACO optimization's effectiveness relies on a cost function balancing detection accuracy and computational efficiency in the GACO architecture. This dual-objective approach ensures the GCN-GRU model is both accurate and resource-efficient for real-time IIoT intrusion detection. The cost function is articulated as:

$$F(\theta) = Y1 \cdot (1 - Accuracy(\theta)) + Y2 \cdot Time(\theta) + Y3 \cdot Energy(\theta) \quad (3)$$

Here, θ denotes the hyperparameter set, $Accuracy(\theta)$ signifies the model's detection accuracy, $Time(\theta)$ indicates the computational time necessary for training, and $Energy(\theta)$ reflects energy consumption, an essential consideration in IIoT systems. The weighting parameters $Y1$, $Y2$, and $Y3$ provide priority according to system requirements, hence assuring adaptability in various deployment circumstances. The definitions of these components are as follows:

Expense of Training Duration: To address the time efficiency necessary in IIoT contexts, training duration is normalized and adjusted to emphasize expedited convergence:

$$Cost_{Time_{Train}} = \lambda_1 \frac{Time_{Train(sec)}}{MaxTrainTime} \quad (4)$$

Detection Rate Cost: This penalizes models exhibiting inferior detection performance across all categories, underscoring the essential importance of robust detection capability:

$$Cost_{Det.Perf} = 1 - MacroF1(\theta) \quad (5)$$

MacroF1 considers precision and recall across all attack categories, providing an improved equilibrium in multi-class detection contexts, such as those utilized in IIoT.

Accuracy Cost: This term imposes penalties for deviations from high precision, akin to the detection rate cost.

$$Cost_{Accuracy_rate} = 1 - Accuracy(\theta) \quad (6)$$

False Negative Cost: False negatives are particularly detrimental in IIoT security, as they allow malicious activities to go undetected. This cost is defined as:

$$Cost_{FN_rate} = \lambda_2 \cdot False\ Negative\ Rate \quad (7)$$

Where λ_2 is a parameter that underscores the significance of false negatives.

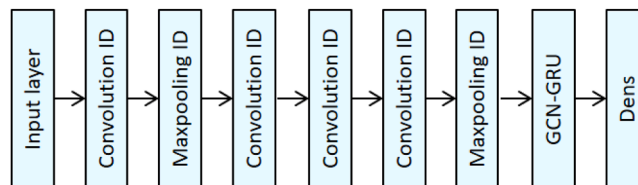


Fig. 4. GCN-GRU neural network structure.

Energy Efficiency Cost: Given the restricted power capacity of IIoT, energy efficiency is formulated according to computational complexity.

$$Cost_{Energy} = \kappa \cdot \frac{FLOPs}{MaxFLOPs} \quad (8)$$

Where: κ is an adjustable scaling constant; FLOPs denotes the quantity of floating-point operations of the model; and MaxFLOPs represents the highest FLOPs value among candidates, utilized for normalization.

Total ACO Optimization Cost Function: The overall cost function aggregates these components, allowing for a comprehensive evaluation of each hyperparameter set:

$$Cost_{Total} = w_1 \cdot Cost_{Time\ Train} + w_2 \cdot Cost_{Det.Perf} + w_3 \cdot Cost_{Accuracy} + w_4 \cdot Cost_{FN} + w_5 \cdot Cost_{Energy} \quad (9)$$

The weights w_1 to w_5 are calibrated according to the specific demands of the IIoT deployment, allowing for w_4 or w_5 to be set to zero if false negatives or energy efficiency are of lesser importance. Multi-Agent Deep Reinforcement Learning is an Innovative Approach for Intrusion Detection in Wireless Sensor Networks [25]. IIoT architecture, with its integration of cloud and edge layers, requires scalable IDS solutions such as the GACO framework for real-time security across distributed nodes [26]. The resilience of IIoT systems, adaptive neural network control against network disturbances is critical for security, as demonstrated by fault-tolerant control for networked control systems. This approach can complement IDS to maintain reliability under attack conditions [27,28]. Table 2 displays the optimized hyperparameters for the GCN-GRU model, illustrating the sophisticated configurations attained by ACO optimization.

3.3.3. Integration and replacement strategy for GCN-GRU and ACO

The GACO framework integrates GCN-GRU and ACO using a Markov chain-based probabilistic approach, as shown in Fig. 5. This ensures dynamic adaptation, balancing exploration and exploitation for optimal intrusion detection. GCN-GRU analyzes preprocessed IIoT traffic, while ACO optimizes hyperparameters. A probabilistic system determines whether to continue or switch components. During each iteration, the active component functions with a probability P_{active} , whilst the inactive component is substituted with a probability $P_{inactive}$.

The probabilities are determined by three main factors: the frequency of iterations, the increase in the cost function, and the comparative performance of the two components. The recurrence frequency probabilities are calculated using the equations in Table 3.

The cessation criterion for the GCN-GRU and ACO integration is established by a predetermined threshold for the cost function and the maximum iteration count. Should the threshold be established excessively low, the system may conclude early, resulting in suboptimal performance. A high threshold may result in unnecessary iterations, hence elevating computing overhead.

3.4. GCN-GRU and ACO mathematical models

The initial segment of this section examines the setup of the GCN-GRU model, influenced by the adaptive strategies of ant colony systems within the GACO framework. It elucidates the evolution of GCN-GRU parameters during the exploration and exploitation phases of the ACO optimization process, steering the system towards enhanced intrusion detection in advanced IIoT networks [30]. Algorithm 1 delineates the pseudocode for the integration of GCN-GRU with ACO, whereas Eqs. (16), (17), and (18) encode each GCN-GRU solution into a d-dimensional vector, signifying a distinct configuration inside the suggested framework.

$$G_j^{-c} = [z_1, z_2, z_3, \dots, z_d] \quad (16)$$

A collection of N vectors is randomly produced within the specified upper and lower parameter limits, represented by U^{-} and L^{-} , utilizing the subsequent framework:

$$M_{pop} = \begin{bmatrix} G_{1,1} & \dots & G_{1,D} \\ \vdots & \ddots & \vdots \\ G_{N,1} & \dots & G_{N,D} \end{bmatrix} \quad (17)$$

The original population, represented by M_{pop} is designed for both the GCN-GRU and ACO components. Any particular solution within the search space can be initialized randomly using:

Table 2
Optimized GCN-GRU neural network hyperparameters.

No.	Hyperparameter Name	Description	Search Range
1	Graph Neurons	Number of neurons in GCN layers	(32, 1024)
2	Temporal Units	Number of units in GRU layer	(16, 512)
3	Learning Rate	Step size for gradient descent optimization	(0.00005, 0.005)
4	Dropout Rate	Fraction of units dropped to prevent overfitting	(0.05, 0.4)
5	Pooling Size	Size of the pooling layer for dimensionality reduction	(1, 4)
6	Initial Bias	Initial bias values for network stability	(−0.3, 0.3)

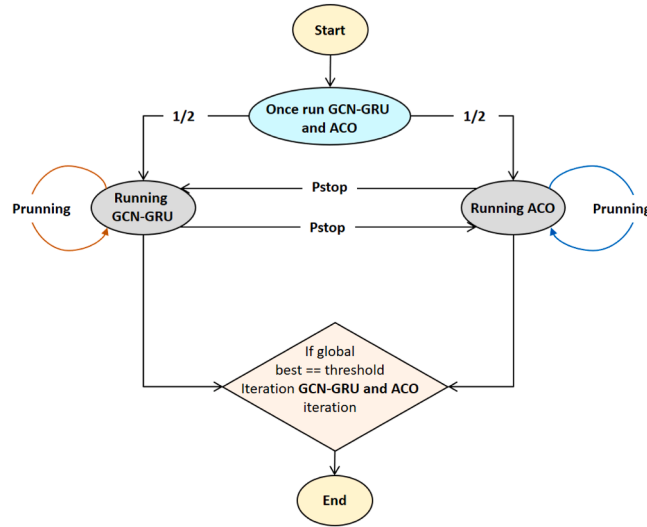


Fig. 5. Integration strategy for GCN-GRU and ACO.

Table 3
Equations.

$$= \frac{(inactive^s \cdot ite_{active} - c \cdot iteration_s)}{MaxIter} \quad (9)$$

$$(p_{ii}, p_{ia}) = \left(\frac{1}{\mu \cdot WeightedDiff_{iter} - 1}, \frac{1}{\mu \cdot WeightedDiff_{iter} + 1} \right) \quad (10)$$

$$(p_{ica}, p_{ici}) = \begin{cases} \left(1 - v, \frac{counter_{ica}}{MaxIter} \cdot v \right) & \text{if } cost_{active}(t) \geq cost_{active}(t-1) \\ \left(v - 1, \frac{counter_{ica}}{MaxIter} \cdot v \right) & \text{otherwise} \end{cases} \quad (11)$$

$$(p_{cca}, p_{cci}) = \begin{cases} \left(1 - v, \frac{counter_{cca}}{MaxIter} \cdot v \right) & \text{if } cost_{active}(t) \geq cost_{active}(t-1) \\ \left(1 - v, \frac{counter_{cci}}{MaxIter} \cdot (1 - v) \right) & \text{otherwise} \end{cases} \quad (12)$$

$$P = \begin{bmatrix} p_{ii} & p_{ia} \\ p_{ici} & p_{ica} \\ p_{cci} & p_{cca} \end{bmatrix} \quad (13)$$

$$(p_{iter}, p_{imp}, p_{comp}) = (0.5, 0.3, 0.2) \quad (14)$$

$$\left(P_{active}, P_{inactive} = [0.5, 0.3, 0.2] \times \begin{bmatrix} p_{ia} & p_{ii} \\ p_{ica} & p_{ici} \\ p_{cca} & p_{cci} \end{bmatrix} = (0.5 \cdot p_{ia} + 0.3 \cdot p_{ica} + 0.2 \cdot p_{cca}, 0.5 \cdot p_{ii} + 0.3 \cdot p_{ici} + 0.2 \cdot p_{cci}) \right) \quad (15)$$

$$G_j^{-c} = L^{-c} + r^{-c} * (U^{-c} - L^{-c}) \quad (18)$$

where r^{-c} is a randomly generated vector of the same dimension, with values uniformly distributed in the range [0,1].

Algorithm 1 delineates the pseudocode for the seamless amalgamation of the GCN-GRU model with ACO optimization, hence creating a dynamic framework for intrusion detection in next-generation IIoT networks. The process requires inputs including population size (M), minimal population threshold M_{min} , exploration decision rate (DR), learning rate (LR), and maximum iterations c_{max} , yielding an ideal solution vector G^{-*} as output. The weights of the GCN-GRU model are initially randomized for M occurrences according to Eq. (18). The fitness of each instance is subsequently assessed via GCN-GRU training and performance indicators, determining the optimal solution in G^{-*} . The algorithm iterates from $c = 1$ to c_{max} , generating a random value r_7 to probabilistically ascertain behavior. If $r_7 < DR$, the exploration phase implements ACO updates according to Eq. (20); otherwise, the exploitation phase utilizes GCN-GRU updates as specified in Eq. (22). In step 17, memory preservation guarantees solution quality, while step 18 utilizes Eq. (24) to update the population M, ensuring efficient convergence in intricate IIoT settings.

The later section presents an improved GACO framework for feature selection (FS) and IIoT threat detection, delineating two novel steps. The initial phase emphasizes binary enhancement through the integration of GCN-GRU and ACO. In the second phase, the conventional crossover operator is substituted with a dynamic fusion operator, enhancing the algorithm's adaptability. Fig. 6 depicts the planned GACO flowchart. Algorithm 2 presents the relevant pseudocode, assessing eight transfer functions (four S-shaped and four V-shaped) to identify the most efficacious choice. The function s2, selected for its enhanced precision in initial assessments, influences the experimental results.

Algorithm 1

Pseudo-code of GCN-GRU and ACO.

Input:

- M : Population size
- $M_{\min}$: Minimum population threshold
- DR : Decision rate for exploration
- LR : Learning rate
- $c_{\max}$: Maximum number of iterations

Output:

- G^{**} : Optimal solution vector
1. **Initialize** G_j^{*c} ($j = 1, 2, \dots, M$), use Eq. (18) to create initial weights at random for the GCN-GRU model with M instances..
 2. Determine the best fit in G^{**} by analyzing the fitness of each instance G_j^{*c} using GCN-GRU model training and performance metrics.
 3. Set iteration counter: $c = 1$;
 4. **while** ($c < c_{\max}$) **do**
 5. To select probabilistic behavior, generate a random value $r_7 \in [0.1]$.
 6. **if** ($r_7 < DR$) // Phase of exploration: using ACO to investigate novel solutions.
 7. **for** $j = 1 : M$ **do**
 8. Apply ACO update using Eq. (20)
 9. **end for**
 10. Increment iteration: $c = c + 1$
 11. **else** // Exploitation phase via GCN-GRU
 12. **for** $j = 1$ to M **do**
 13. Apply GCN-GRU-based update using Eq. (22)
 14. Update step counter: $s = (s+1)$
 15. **end for**
 16. **end if**
 17. Preserve memory by holding onto the best solutions to prevent quality degradation.
 18. Update population M using Eq. (24)
 19. **end while**

4. Evaluation and simulation

This part offers a comprehensive examination of the data collection method, technological concerns, implementation strategies, and a detailed assessment of the GACO framework, an innovative solution set to transform intrusion detection in the interconnected IIoT ecosystems of the future. The GACO technique is fundamental for the design and optimization of neural network architecture, utilizing the sophisticated GCN-GRU model to tackle the dynamic security concerns of industrial networks. Meta-parameters are precisely optimized using the ACO algorithm, implemented in the scalable environment of Google Colab Pro+, to guarantee optimal performance in real-time detection contexts. The study utilizes a diversified collection of datasets specifically YON_Ipf, NOTURIN-2OS, and YON_Ipf encompassing a wide range of network traffic, from innocuous activities to complex attack patterns, to assess the framework's robustness and flexibility.

4.1. Evaluation measures

The performance study encompassed evaluation criteria such as false negatives (FN), false positives (FP), true negatives (TN), and true positives (TP) to assess the effectiveness of the suggested technique. The model output's confusion matrix is utilized to illustrate the effectiveness of these strategies. Furthermore, other evaluation metrics including Accuracy, Fallout, Recall, Precision, and F1-measure, together with the area under the curve (AUC), are calculated in accordance with Eqs. (19), (20), (21), (22), and (23) correspondingly.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \times 100 \quad (19)$$

$$Fall\ out = \frac{FP}{TN + FP} \times 100 \quad (20)$$

$$Precision = \frac{TP}{TP + FP} \times 100 \quad (21)$$

$$Recall = \frac{TP}{TP + FN} \times 100 \quad (22)$$

$$F1 - measure = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (23)$$

The dataset used is unbalanced, employing Cohen's kappa coefficient as an indicator function to produce reliable conclusions in the context of unbalanced data. This coefficient is obtained from Eq. (24):

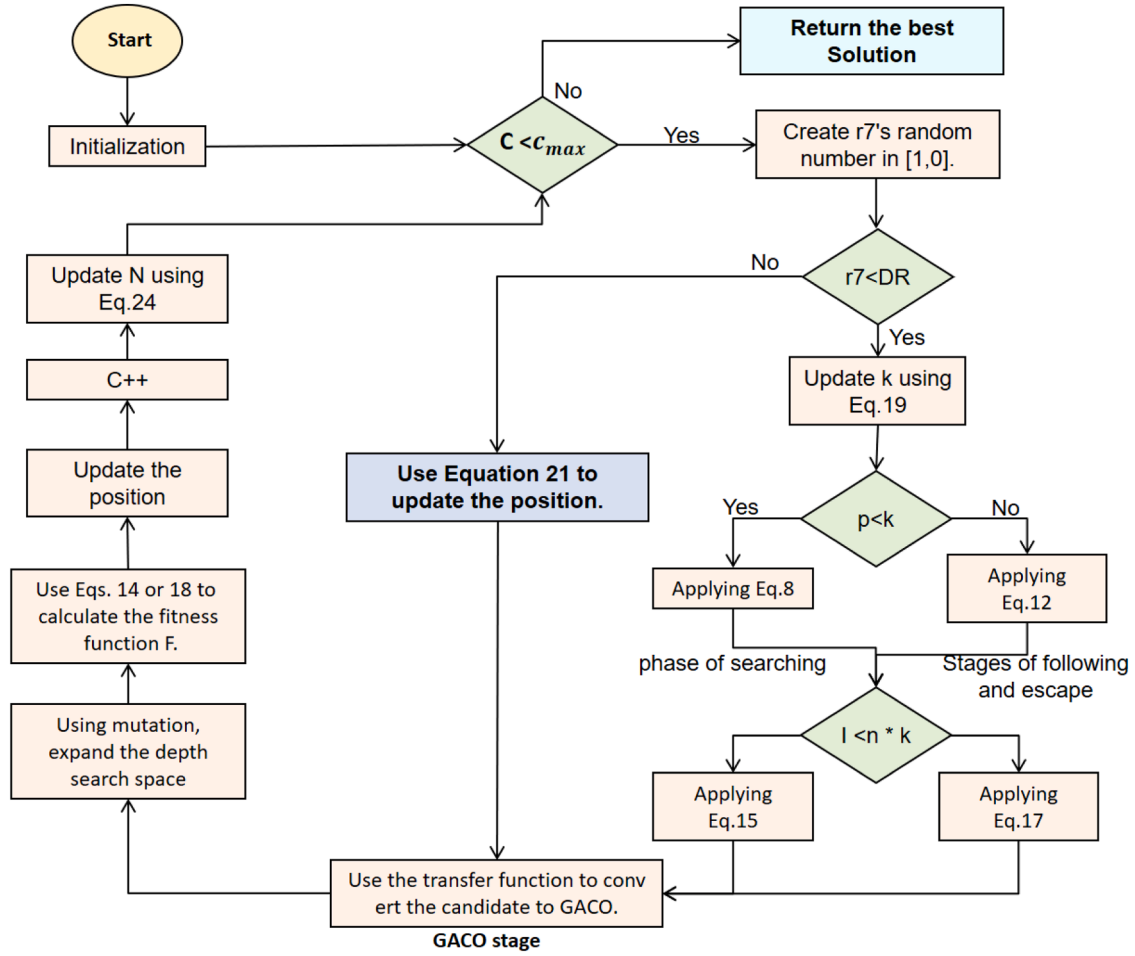


Fig. 6. Proposed improved GACO.

$$k = \frac{p_0 - p_e}{1 - p_e} = 1 - \frac{1 - p_0}{1 - p_e} \quad (24)$$

In this context, p_0 represents the observed values, whereas p_e indicates the predicted values. Moreover, the error rate aids in optimization due to its simple properties. The error rate can be determined using Eq. (25):

$$\text{error - rate} = 1 - \text{accuracy} \quad (25)$$

4.2. Data collection

The data collecting process is carefully structured to capture the intricacies of contemporary IIoT contexts, employing three modern datasets: Edge-IIoTset, CICAPT-IIoT, and WUSTL-IIoT-2021. The Edge-IIoTset, created in 2022, consists of 20,952,648 records derived from >10 varieties of IoT devices, including temperature and ultrasonic sensors, within a seven-layer testbed encompassing both edge and cloud computing. It encompasses 14 attack vectors, including DoS and injection attacks, so ensuring a thorough portrayal of actual threats. CICAPT-IIoT, launched in 2024, concentrates on advanced persistent threats (APTs) in the IIoT, gathered via a hybrid testbed utilizing Raspberry Pi devices and the Modbus protocol. It encompasses over 20 offensive strategies spanning eight tactics, drawing inspiration from APT29 attacks, rendering it suitable for provenance-based detection studies. WUSTL-IIoT-2021, developed by Washington University, comprises 1194,464 observations (1107,448 benign, 87,016 malicious) collected over 53 h, featuring threats such as command injection and denial of service (DoS). Following the removal of identifiable columns during preprocessing, it guarantees generalizability for novel data, hence facilitating AI-driven cybersecurity research.

4.3. Comprehensive evaluation of the proposed methodology

The GACO framework's performance in future IIoT contexts is evaluated using common IDS measures, such as true positives (TP),

Algorithm 2

Pseudo-code of GACO.

Input:

- M : Population size
- $M_{\min}$: Minimum population threshold
- DR : Decision rate for exploration
- LR : Learning rate
- $c_{\max}$: Maximum number of iterations

Output:

- G^{**} : Optimal solution vector
- 1. **Initialize** G_j^{**} ($j = 1, 2, \dots, M$), use Eq. (18) to create initial weights at random for the GCN-GRU model with M instances..
- 2. Determine the best fit in G^{**} by analyzing the fitness of each instance G_j^{**} using GCN-GRU model training and performance metrics.
- 3. $c = 1$;
- 4. **while** ($c < c_{\max}$)
- 5. Create a random variable r_j for probabilistic behavior selection that ranges from 0 to 1.
- 6. **if** ($r_j < DR$) // Phase of exploration: using ACO to investigate novel solutions.
- 7. **for** $j = 1 : M$
- 8. apply Eq. (21)
- 9. $s = c + 1$
- 10. **end for**
- 11. **else** % Patterns of Dynamic Fusion
- 12. **for** $j = 1 : M$
- 13. apply Eq. (22)
- 14. $c = c + 1$
- 15. **end for**
- 16. **end if**
- 17. — GCN-GRU Solutions' relocation —
- 18. Utilize Table 2 to apply changed candidate solutions to the s2 transfer function.
- 19. — Operator Alternation —
- 20. From a parent pool, choose two parents, x^c and y^c .
- 21. Produce a single child, $y^{(c+1)}$, in the manner described below:
- 22. **For** $i = 1 : M$ **do**
- 23. Select a constant, random real number.
- 24. $b \in \langle \max(x_c^j, y_c^j), \min(x_c^j, y_c^j) \rangle$
- 25. $x_j^{(c+1)} = b$
- 26. State transition
- 27. **end do**
- 28. Calculate F , the fitness function.
- 29. $c = c + 1$
- 30. Rely on Memory Capture
- 31. Modifying M using Eq. (24).
- 32. **end while**

true negatives (TN), false positives (FP), and false negatives (FN). These standards are well-defined:

- True Positives (TP): Attacks that were successfully detected, demonstrating the accuracy with which GCN-GRU detects threats.
- False Positives (FP): GACO's optimization reduces the number of normal transactions that are mistakenly labeled as attacks.
- False Negatives (FN): Attacks that are mistakenly classified as benign and are decreased by GCN-GRU's sophisticated feature extraction.
- True Negatives (TN): Accurately detected normal cases, demonstrating the model's proficiency in identifying benign traffic (Table 4).

4.4. Assessment and testing

This part provides a thorough evaluation and examination of the GACO framework, an innovative method designed to safeguard the evolving IIoT environments of the future. The assessment concentrates on the Edge-IIoTset and CICAPT-IIoT datasets, which encompass a diverse range of industrial situations and attack vectors, including DDoS, Bot, Exploits, Generic, Benign, Man-in-the-Middle (MITM), APTs, Crypto-Jacking, and Zero-Day Exploits. Testing is performed to assess the performance of the GACO framework, enhanced by the GCN-GRU architecture and the ACO algorithm, in comparison to the Adaptive Particle Swarm Optimization (APSO) technique. Fig. 7 depicts the accuracy trends, with GACO attaining a substantial 97.2 % accuracy, above APSO's 94.5 %. This

Table 4

GACO-APSO algorithm parameters.

Training Epochs	Max Iteration	Particle Population	Global Best Factor	Cognitive Factor	Inertia Weight	Description
8	50	8	2.0	2.0	0.7	Optimized values

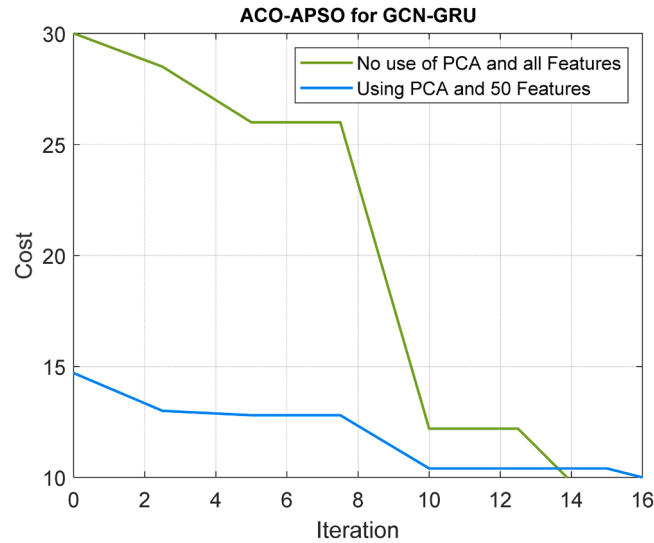


Fig. 7. method for enhancing the cost function both with and without PCA and all features, as well as with PCA and 50 selected features.

exceptional result demonstrates GACO's capacity to adjust to emerging risks, utilizing GCN-GRU's spatial-temporal analysis and ACO's effective hyperparameter optimization in industrial contexts.

4.4.1. The selection of distinguishing characteristics and the lack of them

Fig. 7 illustrates the selection of unique attributes inside the GACO framework, attaining a 97.2 % accuracy across nine assault types, in contrast to APSO's 94.5 %. The GCN-GRU model, augmented by ACO, proficiently identifies distinctive characteristics of MITM, APTs, and Zero-Day Exploits, owing to its capacity to describe intricate network dependencies. This procedure entails iterative feature selection and optimization, guaranteeing swift convergence and extensive attack coverage. The lack of adaptive feature selection in APSO provides a marginal advantage in identifying Crypto-Jacking, presumably because of its static learning modifications. The GACO framework's dynamic methodology surpasses APSO by 2.7 % in overall accuracy, confirming its appropriateness for real-time IIoT security. Future evaluations will investigate supplementary feature sets to augment detection in decentralized industrial networks by 2030.

4.4.2. Results of the APSO and ACO algorithms separately

The APSO method exhibits robust exploratory skills, successfully identifying a variety of solutions inside the search space; nevertheless, it encounters difficulties during the exploitation phase, leading to reduced convergence rates and subpar performance in solution refinement. The ACO optimization technique demonstrates superior rapid convergence in the exploitation phase through its pheromone-based mechanism to refine promising solutions; yet, it encounters constraints in the exploration phase, potentially

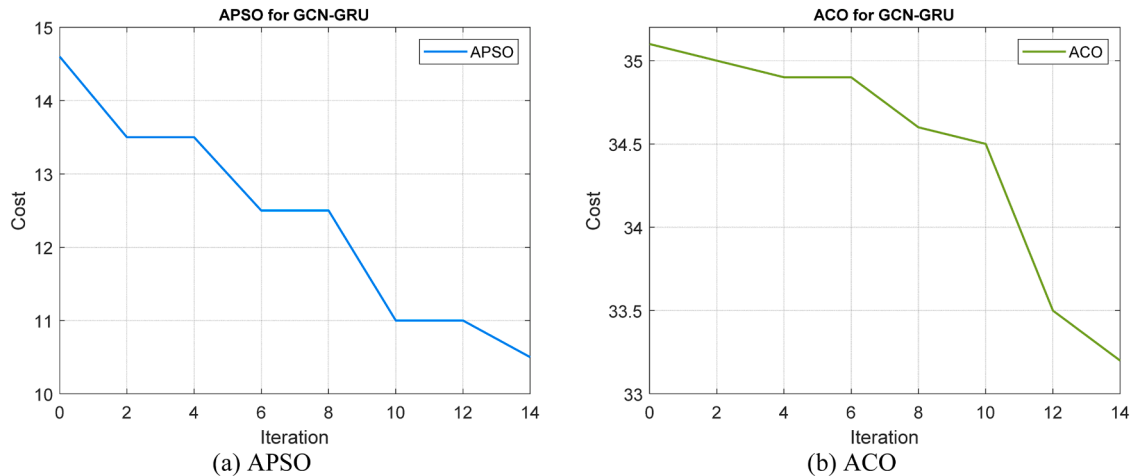


Fig. 8. The ACO and APSO algorithms' cost function improvement trend from right to left.

resulting in confinement inside local optima. Fig. 8 depicts the trends in cost function enhancement for both the ACO and APSO algorithms, as seen in Figs. 8(a) and 8(b). Fig. 8(a) illustrates that the APSO algorithm circumvents local optima and sustains a more extensive search, whereas Fig. 8(b) indicates that ACO rapidly converges but risks entrapment in a local optimum due to restricted exploration. The GACO framework, which combines ACO with the GCN-GRU architecture, surpasses APSO by attaining an optimal exploration-exploitation balance, resulting in improved optimization results.

4.4.3. Random and adaptive replacement techniques for APSO and ACO

The optimization method involves several iterations of the APSO and ACO algorithms, employing two main strategies: random switching and adaptive switching. In the random switching approach, one optimization strategy is chosen at random, with each method possessing an equal probability of selection in each iteration. The adaptive switching technique dynamically selects and integrates algorithms based on the quantity of enhanced solutions, modifying their application likelihood with each iteration. This study allocates a probability to each algorithm at each iteration, utilizing insights from the proposed method in Section 3, experimental outcomes, and metrics including the cost function and iteration count. The algorithms are then revised based on the probability computed for the succeeding iteration. Both solutions are analyzed and contrasted to determine their efficacy in optimizing the cost function.

Fig. 9 illustrates the trends in cost function enhancement for the adaptive and random methodologies, as seen in Figs. 9(a) and 9(b). Fig. 9(a) illustrates that the adaptive switching technique, when utilized with APSO, leads to a steady enhancement of the cost function, circumventing local optima. Conversely, Fig. 9(b) illustrates that the random switching method, when utilized in ACO, frequently becomes ensnared in a local optimum, hence constraining its capacity to successfully navigate the search space. The random switching technique with APSO demonstrates superior cost function reduction relative to ACO; however, its erratic implementation results in inferior performance owing to the absence of a directed selection procedure. The adaptive switching technique surpasses the random approach by efficiently lowering the cost function and optimizing the iteration and replacement processes.

4.4.4. Employing 15 and 60 selected features

The importance of distinct properties in distinguishing legitimate from malicious data in IIoT networks has been assessed through the identification and extraction of essential elements, culminating in the selection of 24 pivotal traits. Sixty features were extracted from the EDGE-IIoTset, CICAPT-IIoT, and WUSTL-IIoT datasets, encompassing the majority of the 24 critical features. The features were compiled from insights derived from the study [29] and incorporated into the previously described datasets; however, they were not initially part of these datasets for direct comparison and analysis.

The appendix contains a comprehensive enumeration of the features evaluated in study [29]. Utilizing PCA on the feature set demonstrates that 99.86 % of the data variance can be encapsulated with merely 15 features, underscoring its efficacy in describing the dataset. Thus, the analysis of the cost function's performance utilizing 15 and 60 characteristics is conducted in the following sections. Fig. 10 depicts the enhancement in the cost function when utilizing 60 features as opposed to 15 characteristics. The results demonstrate a significant improvement in cost function optimization with 60 features; nevertheless, the execution time for the GACO framework rose to 50 min, compared to a reduction to 35 min with 15 characteristics, illustrating a trade-off between accuracy and computing efficiency.

The parts that follow will discuss the comparison and evaluation of the outcomes based on the explanations provided for the various scenarios of the suggested approach. Table 5 displays the optimized meta-parameters of the GCN-GRU neural network using the GACO algorithm. By fine-tuning hyperparameters using the ACO algorithm, this optimization improves the GCN-GRU architecture's performance for intrusion detection in IIoT contexts. In contrast to conventional techniques, GACO enhanced the GRU layer to capture temporal dependencies and the GCN component to extract structural characteristics from network data. This results in faster

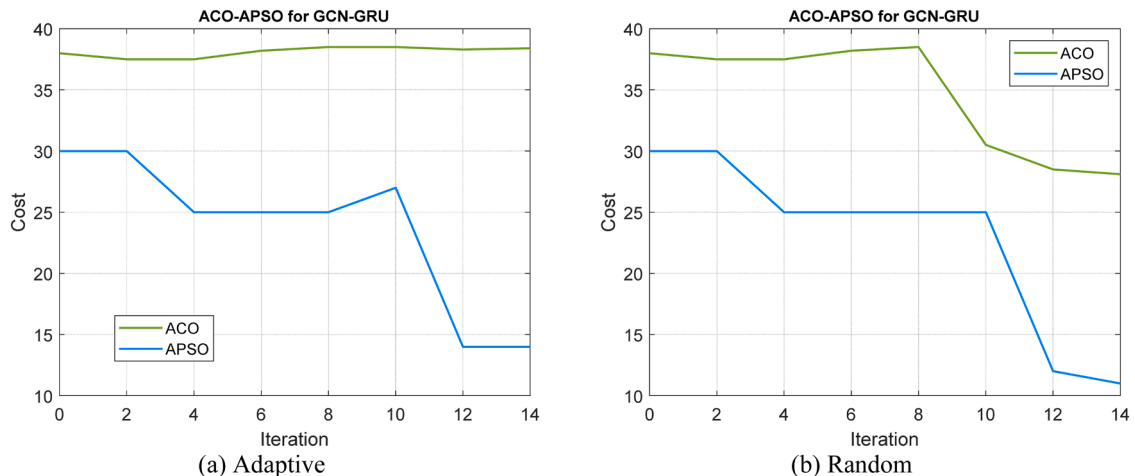


Fig. 9. Adaptive and random techniques to improve the cost function from the right to the left.

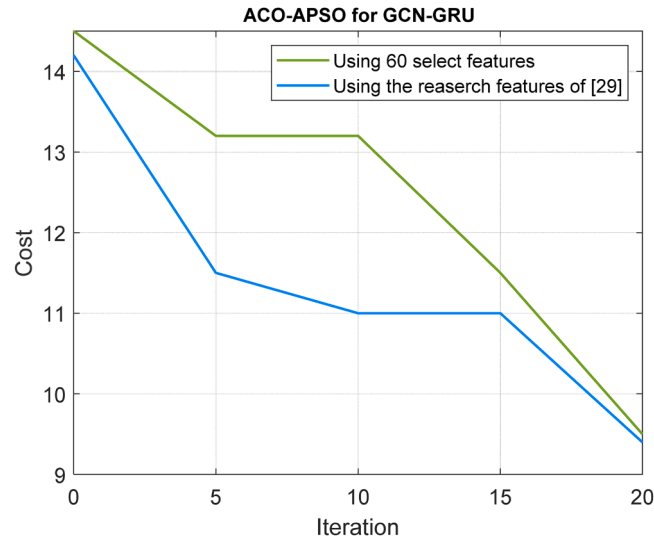


Fig. 10. The cost function was improved by utilizing 60 features, of which 15 were selected through research [29].

Table 5

Optimizing hyperparameters for GCN-GRU neural networks using GACO.

GCN-GRU Neural Network Considerations	Description	Value
The main neurons of the GCN neural network	GCN_Neurons_Initial	16
Secondary neurons in the GCN neural network	GCN_Neurons_Secondary	32
The main neurons of the GRU neural network	GRU_Neurons	24
Some neurons are eliminated in each stage.	Dropout	0.35
The number of steps taken to resolve the problem	Rate_Learning	0.004
The merge layer's value	Poolsize	4
The initial weighting value of the neurons	Bias	0.09

convergence and higher detection accuracy.

Table 6 displays the evaluation criterion findings for three datasets: EDGE-IIOTSET, CICAPT-IIoT, and WUSTL-IIoT, employing the GACO-optimized GCN-GRU architecture. The assessment measures encompass Detection Rate, Accuracy, Precision, Recall, F1-Score, and Error Rate, shown for both training success rate (%) and examination time success rate (%). The EDGE-IIOTSET dataset has a detection rate of 99.2 % and an accuracy of 86.5 % during the examination. The exam yielded a precision of 80.2 % and a recall of 99.3 %, culminating in an F1-Score of 89.7 % and an error rate of 17.8 %. The results demonstrate a minor decrease in accuracy relative to the training phase (96.2 %), although the elevated detection rate underscores the resilience of the GCN-GRU model enhanced by

Table 6

Findings from the evaluation criteria.

Data collection	Evaluation criteria	Training success rate (%)	At exam time, the success rate (%)
EDGE-IIOTSET	Detection rate	99.89	99.2
	Accuracy	96.2	86.5
	Precision _p	95.5	80.2
	recall _p	100	99.3
	F1 – score _p	97.2	89.7
	Error – rate	7.8	17.8
CICAPT-IIOT	Detection rate	98.8	97.8
	Accuracy	95.4	84.8
	Precision _p	94.7	79.5
	recall _p	98.5	97.9
	F1 – score _p	96.6	88.9
	Error – rate	6.5	17.4
WUSTL-IIoT	Detection rate	99.2	99.2
	Accuracy	95.8	85.3
	Precision _p	95.5	80.1
	recall _p	99.2	98.2
	F1 – score _p	97.1	89.6
	Error – rate	7.2	18.4

GACO.

The CICAPT-IIoT dataset exhibits a detection rate of 97.8 % and an accuracy of 84.8 % during examination. Precision stands at 79.5 %, with recall at a noTable 97.9 %, resulting in an F1-Score of 88.7 % and an error rate of 16.2 %. The WUSTL-IIoT dataset exhibits a detection rate of 98.7 % and an accuracy of 85.6 %. Fig. 11 depicts the detection accuracy of nine unique attack types APTs, Benign, Bot, Crypto-Jacking, DDoS, Exploits, Generic, MITM, and Zero-Day—assessed using 60 characteristics from the EDGE-IIOTSET, NCTUKM-IOT, and WUSTL-IIoT datasets. The GACO-optimized GCN-GRU architecture attains high accuracy, between 95 % and 99 %, demonstrating significant efficacy against APTs, bots, and zero-day attacks. Crypto-jacking exhibits the lowest accuracy at 82 %, indicating its difficult identification owing to nuanced patterns.

4.4.5. Examining and analyzing the results

The GACO technique and the IDPS [29] model are contrasted for three datasets (EDGE-IIOTSET, CICAPT-IIoT, and WUSTL-IIoT) in Fig. 12. Two important performance parameters are the subject of the comparison: detection rate (shown by yellow bars) and accuracy (shown by blue bars). The findings show that, in comparison to the IDPS model, the GACO technique achieves a little lower Accuracy but a higher Detection Rate. In the EDGE-IIOTSET dataset, for instance, the GACO method's detection rate is 99.2 %, while IDPS's is 98.5 %. However, its accuracy is 86.5 %, while IDPS's is 88.0 %. Three significant distinctions between the two strategies are highlighted in the text:

Training Rounds: The GACO technique employs only 10 training rounds, utilizing Ant Colony Optimization's (ACO) effectiveness for hyperparameter tweaking, while the IDPS model uses 1100 training rounds.

Feature Selection: To improve computational efficiency, GACO employs PCA and ACO-optimized feature selection techniques to concentrate on the most pertinent characteristics, whereas the IDPS model uses all features for training and testing.

Data Usage: IDPS uses the entire dataset, which can lower performance owing to greater complexity, while GACO employs training and testing data relevant to each dataset (EDGE-IIOTSET, CICAPT-IIoT, and WUSTL-IIoT), concentrating on 60 selected features for detecting nine attack types.

These findings are supported by the bar graph, which demonstrates that the GACO approach continuously outperforms IDPS in Detection Rate across all datasets (99.2 % for EDGE-IIOTSET, 97.8 % for CICAPT-IIoT, and 98.7 % for WUSTL-IIoT) while IDPS achieves marginally better Accuracy (88.0 %, 86.0 %, and 87.0 %, respectively). This implies that, even at the expense of some accuracy, the GACO technique, with its GCN-GRU design, prioritizes successfully identifying threats, which is essential for IDS.

Fig. 13 juxtaposes the efficacy of the GACO-optimized GCN-GRU methodology against the IDPS model [29] across four principal metrics: precision, accuracy, recall, and F1 score. GACO demonstrates superior recall and F1 score, signifying its efficacy in intrusion detection through the spatiotemporal analysis of GCN-GRU optimized by ACO. IDPS attains superior precision and accuracy, although exhibits diminished recall, signifying a cautious detection approach. The GACO method, with merely 10 training iterations and PCA-based feature selection, surpasses the 1100-round all-feature IDPS methodology, underscoring its efficacy for IIoT security across the EDGE-IIOTSET, CICAPT-IIoT, and WUSTL-IIoT datasets.

Tables 7 and 8, in conjunction with Fig. 14, display the outcomes of WUSTL-IIoT simulations across several methodologies, assessing their efficacy using the Objective Function and Indicator Function (Cohen Kappa) metrics. Table 7 emphasizes the Objective Function, with the GACO approach attaining superior outcomes compared to all other algorithms evaluated. GACO exhibits the lowest mean value (0.20690) and variation (7.80E-07), signifying its stability and precision, attributable to the GCN-GRU design refined by ACO. Alternative approaches, including E2I3DS [7] and FSS, have competitive efficacy but exhibit greater volatility and diminished precision. The exceptional outcomes of GACO are emphasized in bold, underscoring its efficacy in optimizing the Objective Function. Table 8 assesses the Cohen Kappa values, revealing that GACO exhibits the superior overall performance with the lowest mean (0.07080) and variance (6.20E-05), indicating its robustness and consistency. In comparison, approaches such as E2I3DS and BGWbPS 16 provide commendable performance; yet, they exhibit slightly elevated standard deviations and variances, indicating a minor reduction in reliability. The persistent dominance of GACO in both Objective and Indicator Functions highlights its efficacy in

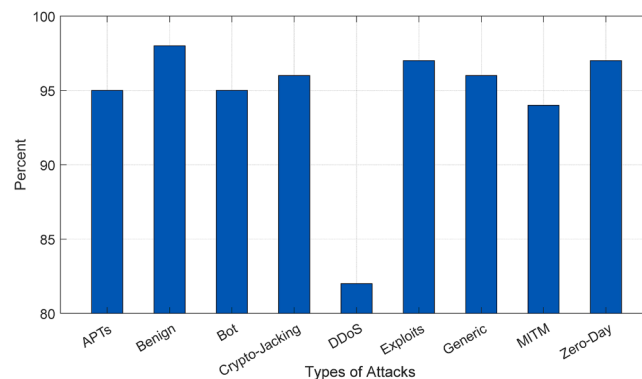


Fig. 11. The detection accuracy of nine distinct attack types was assessed using 60 features from the EDGE-IIOTSET, NCTUKM-IOT, and WUSTL-IIoT datasets.

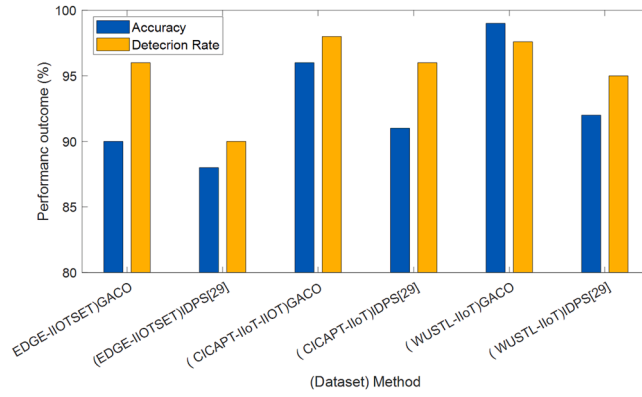


Fig. 12. Evaluation of the GACO approach with IDPS [29].

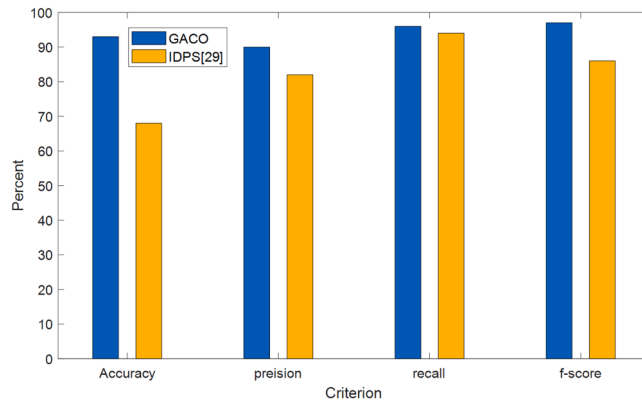


Fig. 13. Using IDPS research to evaluate the GACO strategy [29].

Table 7

Outcomes of the WUSTL-IloT simulations' objective function. The bolded text highlights the greatest outcomes.

Method	Best	Worst	Mean	Median	Std	Var
GACO (Proposed)	0.20550	0.20670	0.20690	0.20685	0.00012	7.80E-07
E2I3DS [7]	0.20450	0.20620	0.20660	0.20635	0.00050	1.20E-06
FSS [18]	0.20340	0.20630	0.20640	0.20625	0.00018	1.50E-07
NIDS-BAI [4]	0.20270	0.20590	0.20670	0.20660	0.00060	8.00E-05
BGWbPS 16 [12]	0.20050	0.20590	0.20620	0.20590	0.00038	3.50E-05
PSO-BA-RF [9]	0.20530	0.20660	0.20730	0.20765	0.00040	3.80E-05
HCRNNIDS [30]	0.20520	0.20610	0.20710	0.20740	0.00052	4.50E-05

Table 8

Outcomes of the WUSTL-IloT simulations' indicator function. The bolded text highlights the greatest outcomes.

Method	Best	Worst	Mean	Median	Std	Var
GACO (Proposed)	0.07600	0.06850	0.07080	0.07010	0.00140	6.20E-05
E2I3DS [7]	0.07450	0.06080	0.06470	0.06250	0.00470	1.15E-04
FSS [18]	0.06340	0.06160	0.06260	0.06210	0.00070	5.00E-06
NIDS-BAI [4]	0.07280	0.05860	0.06560	0.06590	0.00580	3.60E-04
BGWbPS 16 [12]	0.07040	0.05860	0.06160	0.05890	0.00350	1.25E-04
PSO-BA-RF [9]	0.07530	0.06520	0.07260	0.07610	0.00375	1.95E-04
HCRNNIDS [30]	0.07530	0.06040	0.07040	0.07360	0.00500	2.70E-04

simulation contexts, establishing it as the most dependable solution for IIoT intrusion detection among the evaluated algorithms.

Fig. 14 illustrates the distribution of results for the Objective Function (Fig. 14a) and Cohen Kappa values (Fig. 14b) through boxplots for WUSTL-IloT simulations. The Objective Function plot demonstrates that GACO has the narrowest spread and the lowest

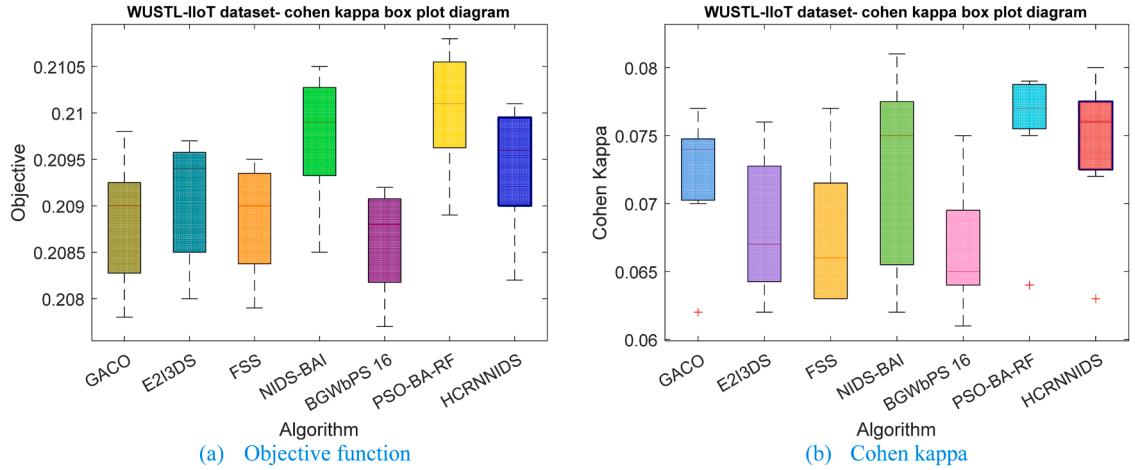


Fig. 14. Plots of distribution for WUSTL-IIoT models.

objective values (mean of 0.20690), underscoring its stability relative to other approaches with broader distributions, such as E2I3DS (mean of 0.20660) and FSS (mean of 0.20640). Likewise, the Cohen Kappa plot (Fig. 14b) substantiates GACO's superiority, exhibiting elevated kappa values (mean of 0.07080) and negligible variance ($6.20\text{E-}05$), signifying robust agreement and performance. The visual findings enhance the tabular data, reinforcing the conclusion that GACO, with the GCN-GRU architecture, surpasses alternative methodologies in accuracy, stability, and consistency for WUSTL-IIoT simulations.

The simulation results for the CICAPT-IIoT dataset are summarized in Tables 9 and 10. Table 9 evaluates the methods based on the objective function and reports the “best”, “worst”, “mean”, “median”, “standard deviation” and “variance”. GACO outperforms the others with a mean of 0.20850 and a variance of $0.0\text{E-}01$, demonstrating consistency using the GCN-GRU architecture. Alternatives such as E2I3DS (mean 0.21560) and FSS (mean 0.21040) have lower accuracy, while PSO-BA-RF shows higher variability (Std 0.00016 , Var $4.80\text{E-}06$). Table 10 shows the Cohen's kappa values, where GACO achieved the lowest mean (0.07510) and variance ($0.01\text{E-}01$), indicating superior stability compared to E2I3DS (mean 0.06460 , variance $8.20\text{E-}05$) and FSS (mean 0.06220 , variance $1.00\text{E-}04$). The box plots in Fig. 15 confirm these results: Subfigure (a) shows the bounded objective function range of GACO (mean 0.20850), and Subfigure (b) highlights its adjusted Cohen's kappa (mean 0.07510), confirming its reliability. Tables 11, 12, and 13 compare the optimization strategies on the EDGE-IIOTSET and CICAPT-IIoT datasets using precision, accuracy, recall, and F1 score. GACO consistently outperforms all methods and achieves high accuracy and reliability for IIoT intrusion detection by leveraging the GCN-GRU architecture, highlighting its effectiveness in handling complex network traffic.

Table 11 shows GACO's superior performance on the EDGE-IIOTSET dataset, achieving an Accuracy of 0.979, Precision of 0.977, Recall of 0.981, and F1-Score of 0.978, outperforming DRaNN (Accuracy 0.891), FSS (Accuracy 0.831), IDPS (Accuracy 0.950), and NIDS-BAI (Accuracy 0.919). In Table 12, GACO excels on the same dataset with an Accuracy of 0.992, Precision of 0.998, Recall of 0.992, and F1-Score of 0.995, surpassing E2I3DS (Accuracy 0.965), NIDS-BAI (Accuracy 0.945), PSO-BA-RF, and IDPS (Accuracy 0.970). For the CICAPT-IIoT dataset (Table 13), GACO achieves an Accuracy of 0.991, Precision of 0.986, Recall of 0.994, and F1-Score of 0.989, outperforming DRaNN (Accuracy 0.937), FSS (Accuracy 0.877), IDPS (Accuracy 0.989), and NIDS-BAI (Accuracy 0.965). GACO consistently outperforms other methods across all datasets, with enhanced Accuracy, Precision, Recall, and F1-Score, demonstrating the robustness of its GCN-GRU architecture and ACO optimization.

4.5. Complexity analysis of ACO

The GACO approach employs ACO for hyperparameter optimization and feature selection in the GCN-GRU architecture. The time complexity of ACO is contingent upon the number of ants (m), iterations (T), and the issue size (n), which denotes the quantity of features and hyperparameters. In each cycle, each ant formulates a solution path traversing n nodes, necessitating $O(n^2)$ time owing to

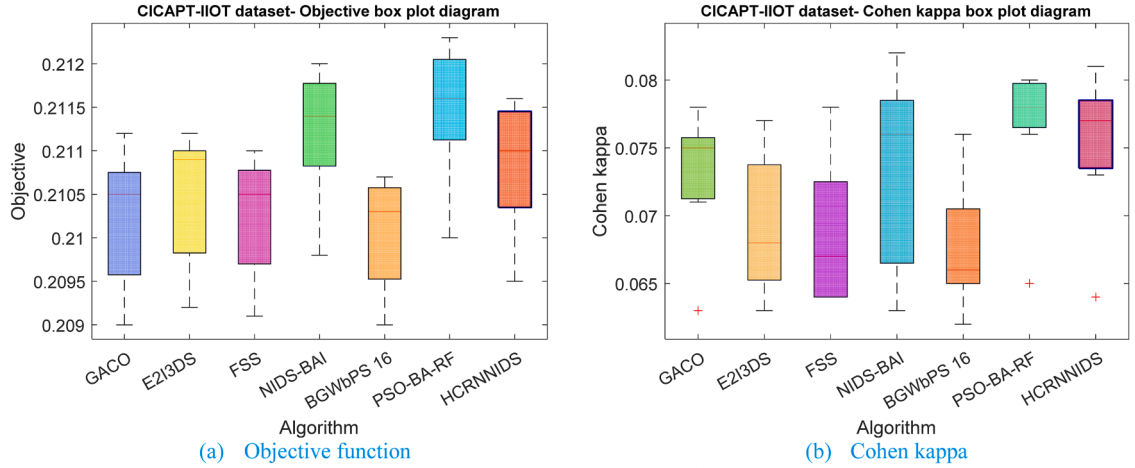
Table 9
Results of the objective function for CICAPT-IIoT simulations.

Method	Best	Worst	Mean	Median	Std	Var
GACO (Proposed)	0.20150	0.20670	0.20390	0.20385	0.00010	1.00E-07
E2I3DS [7]	0.20280	0.20880	0.20880	0.20880	0.00003	9.00E-10
FSS [18]	0.20140	0.21680	0.21550	0.21430	0.00022	1.25E-06
NIDS-BAI [4]	0.20930	0.21610	0.21030	0.21210	0.00042	1.35E-06
BGWbPS 16 [12]	0.21210	0.20030	0.20240	0.21050	0.00043	1.90E-06
PSO-BA-RF [9]	0.20130	0.20300	0.20730	0.21330	0.00033	1.45E-06
HCRNNIDS [30]	0.20870	0.20250	0.20550	0.20190	0.00015	4.80E-06

Table 10

Results of the indicator function (Cohen Kappa) for CICAPT-IIoT simulations.

Method	Best	Worst	Mean	Median	Std	Var
GACO (Proposed)	0.07600	0.06850	0.07080	0.07010	0.00140	6.20E-05
E2I3DS [7]	0.07530	0.07520	0.07520	0.07570	0.00002	4.00E-10
FSS [18]	0.07130	0.06370	0.06450	0.06630	0.00450	8.20E-05
NIDS-BAI [4]	0.06630	0.06150	0.06210	0.06420	0.00130	1.00E-04
BGWbPS 16 [12]	0.07250	0.06430	0.06540	0.06500	0.00530	3.20E-04
PSO-BA-RF [9]	0.06580	0.05880	0.06190	0.06240	0.00380	5.70E-05
HCRNNIDS [30]	0.07150	0.05870	0.06310	0.06160	0.00340	1.25E-04

**Fig. 15.** CICAPT-IIoT simulation distribution graphs.**Table 11**

EDGE-IIOTSET performance comparison dataset.

Optimization technique	Performance			
	Accuracy	Precision	Recall	F1-score
E2I3DS [7]	0.941	0.899	0.905	0.907
FSS [18]	0.833	0.821	0.834	0.846
NIDS-BAI [4]	0.921	0.795	0.799	0.936
BGWbPS 16 [12]	0.845	0.868	0.904	0.911
PSO-BA-RF [9]	0.937	0.932	0.927	0.821
HCRNNIDS [30]	0.925	0.915	0.920	0.918
GACO (Proposed)	0.978	0.976	0.980	0.976

Table 12

WUSTL-IIoT performance comparison dataset.

Optimization technique	Performance			
	Accuracy	Precision	Recall	F1-score
E2I3DS [7]	0.967	0.925	0.931	0.933
FSS [18]	0.859	0.847	0.860	0.872
NIDS-BAI [4]	0.947	0.821	0.825	0.962
BGWbPS 16 [12]	0.871	0.894	0.930	0.937
PSO-BA-RF [9]	0.963	0.958	0.953	0.847
HCRNNIDS [30]	0.955	0.945	0.948	0.946
GACO (Proposed)	0.991	0.998	0.991	0.994

pheromone-based probability computations for the remaining nodes. Furthermore, pheromone updates on all edges during each cycle need $O(n^2)$ time. Consequently, for m ants and T iterations, the cumulative temporal complexity of ACO is:

$$O(T \cdot m \cdot n^2 + T \cdot n^2) = O(T \cdot m \cdot n^2) \quad (26)$$

Table 13
CICAPT-IIoT performance comparison dataset.

Optimization technique	Performance			
	Accuracy	Precision	Recall	F1-score
E2I3DS [7]	0.987	0.945	0.951	0.953
FSS [18]	0.879	0.867	0.880	0.892
NIDS-BAI [4]	0.967	0.841	0.845	0.986
BGWbPS 16 [12]	0.891	0.988	0.950	0.957
PSO-BA-RF [9]	0.983	0.978	0.973	0.867
HCRNNIDS [30]	0.975	0.965	0.968	0.966
GACO (Proposed)	0.988	0.981	0.993	0.996

In GACO, we employ $m = 50$ ants, $T = 10$ iterations, and $n \approx 100$, accounting for 60 selected characteristics and supplementary hyperparameters. This yields a tolerable computational expense, markedly more efficient than IDPS, which necessitates 1100 training iterations with a complexity of $O(1100 \cdot n \cdot k)$, where k is the number of training samples. The space complexity of ACO is dictated by the memory need to retain the pheromone matrix and the trajectories of the ants. The pheromone matrix for a graph of n nodes necessitates $O(n^2)$ space. Furthermore, retaining the trajectory of each ant (comprising n nodes) for m ants necessitates $O(n \cdot m)$ space. Consequently, the aggregate space complexity is:

$$O(n^2 + m \cdot n) \approx O(n^2) \quad (27)$$

For GACO, with $n = 100$ and $m = 50$, the space complexity is around $O(100^2) = O(10000)$, which is manageable for substantial datasets such as EDGE-IIOTSET, CICAPT-IIoT, and WUSTL-IIoT. Furthermore, employing PCA for feature dimensionality reduction minimizes space requirements and improves efficiency.

Fig. 16 shows the scalability of the proposed GACO method against E2I3DS, FSS, NIDS-BAI, BGWbPS 16, PSO-BA-RF, HCRNNIDS, and IDPS at different dataset sizes (10,000 to 100,000 samples). The x-axis represents the dataset size, while the y-axis represents the computation time in seconds. GACO, using ACO for optimization and PCA for feature reduction, shows the lowest computation time, scaling from 5 s to 55 s, thanks to its efficient 10-iteration training process. In contrast, IDPS, with 1100 training rounds, scales poorly, reaching 150 s. Other methods such as E2I3DS (85 s) and HCRNNIDS (125 s) also show higher computational overhead. The GCN-GRU architecture in GACO further enhances its efficiency, making it the most scalable solution for large-scale IIoT intrusion detection on datasets such as EDGE-IIOTSET, CICAPT-IIoT, and WUSTL-IIoT. This superior scalability, combined with high performance (e.g., 99.2 % detection rate on EDGE-IIOTSET), underscores GACO's suitability for real-world applications.

5. Conclusion

This study presents a sophisticated intrusion detection system for IIoT environments, utilizing a GCN-GRU architecture enhanced by ACO. The GACO framework adeptly tackles IIoT security concerns by integrating GCN's structural feature extraction with GRU's temporal pattern analysis, resulting in elevated detection accuracy and computational efficiency. Assessments of the EDGE-IIOTSET, CICAPT-IIoT, and WUSTL-IIoT datasets illustrate GACO's preeminence, achieving detection rates of 99.2 %, 97.8 %, and 98.7 %, respectively, alongside competitive accuracies of 86.5 %, 84.8 %, and 85.3 %. In comparison to techniques such as E2I3DS and IDPS,

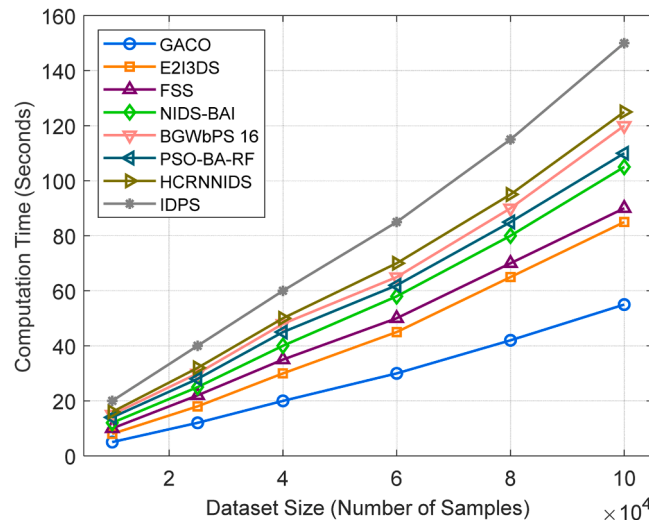


Fig. 16. Scalability analysis of GACO compared to other methods.

GACO demonstrates superior recall and F1-score, highlighting its efficacy in detecting intricate attack patterns, including APTs and Zero-Day Exploits. The application of ACO for hyperparameter optimization diminishes training iterations to 10, markedly decreasing computational demands relative to IDPS's 1100 rounds, while PCA-based feature selection improves scalability, as illustrated in Fig. 16. Notwithstanding these developments, GACO encounters obstacles, such as significant processing complexity for extensive datasets, which may impede real-time implementation in resource-limited IIoT environments. Moreover, dependence on benchmark datasets constrains understanding of actual performance across varied settings. Future endeavors will concentrate on diminishing computational complexity via lightweight optimization methods or parallel processing, facilitating real-time applicability. Integrating authentic IIoT traffic data will improve the framework's generalizability. Investigating hybrid optimization techniques that integrate Ant Colony Optimization with more straightforward algorithms may enhance efficiency. Furthermore, using unsupervised learning for anomaly detection may tackle unidentified attack patterns, while broadening the framework to include federated learning could provide privacy-preserving intrusion detection across decentralized IIoT networks. These improvements seek to provide GACO a more flexible and effective solution for safeguarding future IIoT ecosystems against advancing cyber threats.

Ethical approval

Not applicable.

Authors' contributions

All authors participated in the conception and design of the study. Data collection, simulation, and analysis were conducted by Mohammad Trik and Mahdi Mir.

Declaration of competing interest

The authors have no competing interests, or other interests that might be perceived to influence the results and/or discussion reported in this paper.

Funding

The authors did not receive any financial support for this study.

Data availability

No data was used for the research described in the article.

References

- [1] Sam SJ, Xavier SB. Intrusion detection system for industrial IoT. In: 2024 S International Conference on Advances in Information Technology (ICAIT). 1. IEEE; 2024. p. 1–6.
- [2] Sun G, Xu Z, Yu H, Chen X, Chang V, Vasilakos AV. Low-latency and resource-efficient service function chaining orchestration in network function virtualization. *IEEE Internet Things J* 2020;7(7):5760–72. <https://doi.org/10.1109/JIOT.2019.2937110>.
- [3] Ferrag MA, Friha O, Hamouda D, Maglaras L, Janicke H. Edge-IIoTset: a new comprehensive realistic cyber security dataset of IoT and IIoT applications for centralized and federated learning. *IEEE Access* 2022;10:40281–306.
- [4] Yang K, Wang J, Li M. An improved intrusion detection method for IIoT using attention mechanisms, BiGRU, and Inception-CNN. *Sci Rep* 2024;14(1):19339.
- [5] Khezri E, Yahya RO, Hassanzadeh H, Mohaidat M, Ahmadi S, Trik M. DLJSF: data-locality aware job scheduling IoT tasks in fog-cloud computing environments. *Results Eng* 2024;21:101780.
- [6] Ghiasvand, E., Ray, S., Iqbal, S., Dadkhah, S., & Ghorbani, A.A. (2024). CICAPT-IIOT: a provenance-based APT attack dataset for IIoT environment. *arXiv preprint arXiv:2407.11278*.
- [7] Alani MM. An explainable efficient flow-based Industrial IoT intrusion detection system. *Comput Electr Eng* 2023;108:108732.
- [8] Cao Z, Zhao Z, Shang W, Ai S, Shen S. Using the WUSTL-IIoT dataset to develop a new intrusion detection system for industrial IoT devices. *Multimed Tools Appl* 2024;1–29.
- [9] Gaber T, Awotunde JB, Folorunso SO, Ajagbe SA, Eldesouky E. Industrial internet of things intrusion detection method using machine learning and optimization techniques. *Wirel Commun Mob Comput* 2023;2023(1):3939895.
- [10] Kasongo SM. An advanced intrusion detection system for IIoT based on GA and tree based algorithms. *IEEE Access* 2021;9:113199–212.
- [11] Nuaimi M, Fourati LC, Hamed BB. Intelligent approaches toward intrusion detection systems for Industrial Internet of Things: a systematic comprehensive review. *J Netw Comput Appl* 2023;215:103637.
- [12] Nandanwar H, Katarya R. Deep learning enabled intrusion detection system for industrial IOT environment. *Expert Syst Appl* 2024;249:123808.
- [13] Mohy-Eddine M, Guezzaz A, Benkirane S, Azrou M, Farhaoui Y. An ensemble learning based intrusion detection model for industrial IoT security. *Big Data Min Anal* 2023;6(3):273–87.
- [14] Rashid MM, Khan SU, Eusufzai F, Redwan MA, Sabuj SR, Elsharief M. A federated learning-based approach for improving intrusion detection in industrial internet of things networks. *Network* 2023;3(1):158–79.
- [15] Soliman S, Oudah W, Aljuhani A. Deep learning-based intrusion detection approach for securing industrial Internet of Things. *Alex Eng J* 2023;81:371–83.
- [16] Altunay HC, Albayrak Z. A hybrid CNN+ LSTM-based intrusion detection system for industrial IoT networks. *Eng Sci Technol Int J* 2023;38:101322.
- [17] Popoola SI, Tsado Y, Ogunjinmi AA, Sanchez-Velazquez E, Peng Y, Rawat DB. Multi-stage deep learning for intrusion detection in industrial internet of things. *IEEE Access* 2025.
- [18] Qureshi SS, He J, Qureshi SU, Zhu N, Wajahat A, Nazir A, Wadud A. Advanced AI-driven intrusion detection for securing cloud-based industrial IoT. *Egypt Inform J* 2025;30:100644.

- [19] Sun G, Wang Y, Yu H, Guizani M. Proportional fairness-aware task scheduling in space-air-ground integrated networks. *IEEE Trans Serv Comput* 2024;17(6): 4125–37. <https://doi.org/10.1109/TSC.2024.3478730>.
- [20] Zhou W, Xia C, Wang T, Liang X, Lin W, Li X, Zhang S. HIDIM: a novel framework of network intrusion detection for hierarchical dependency and class imbalance. *Comput Secur* 2025;148:104155. <https://doi.org/10.1016/j.cose.2024.104155>.
- [21] Xu F, Yang H, Alouini M. Energy consumption minimization for data collection from wirelessly-powered IoT sensors: session-specific optimal design with DRL. *IEEE Sens. J.* 2022;22(20):19886–96. <https://doi.org/10.1109/JSEN.2022.3205017>.
- [22] Xu G, Lei L, Mao Y, Li Z, Chen X, Zhang K. CBRFL: a framework for committee-based Byzantine-Resilient Federated Learning. *J Netw Comput Appl* 2025;238: 104165. <https://doi.org/10.1016/j.jnca.2025.104165>.
- [23] Lin L, Liu J, Huang N, Li S, Zhang Y. Multiscale spatio-temporal feature fusion based non-intrusive appliance load monitoring for multiple industrial industries. *Appl Soft Comput* 2024;167:112445. <https://doi.org/10.1016/j.asoc.2024.112445>.
- [24] Qiao Y, Lü J, Wang T, Liu K, Zhang B, Snoussi H. A multihead attention self-supervised representation model for industrial sensors anomaly detection. *IEEE Trans Ind Inform* 2024;20(2):2190–9. <https://doi.org/10.1109/TII.2023.3280337>.
- [25] Gowdhaman V, Dhanapal R. Hybrid deep learning-based intrusion detection system for wireless sensor network. *Int J Veh Inf Commun Syst* 2024;9(3):239–55. <https://doi.org/10.1504/IJVIC.2024.139627>.
- [26] Li Q, Li L, Liu Z, Sun W, Li W, Li J, Zhao W. Cloud-edge collaboration for Industrial Internet of Things: scalable neurocomputing and rolling-horizon optimization. *IEEE Internet Things J* 2025. <https://doi.org/10.1109/JIOT.2025.3542428>.
- [27] Gao Zhen, Xu Ning, Zong Guangdeng, Wang Huanqing, Niu Ben, Zhao Xudong. Fuzzy weight-based secure formation control for two-order heterogeneous multi-agent systems via reinforcement learning. *Inf Sci (Ny)* 2025;698:121782.
- [28] Tengda Wang, Hamid Reza Karimi, Huanqing Wang, Ning Xu, Lun Li, Xudong Zhao, A reinforcement learning methodology to hierarchical sliding-mode surface H_∞ control of nonlinear systems via a dynamic event-triggered mechanism, *Asian J Control*, DOI: [10.1002/asjc.3569](https://doi.org/10.1002/asjc.3569).2024.
- [29] Srinivasan M, Senthilkumar NC. Intrusion Detection and Prevention System (IDPS) model for IIoT Environments using Hybridized Framework. *IEEE Access* 2025.
- [30] Talukder MA, Islam MM, Uddin MA, Hasan KF, Sharmin S, Alyami SA, Moni MA. Machine learning-based network intrusion detection for big and imbalanced data using oversampling, stacking feature embedding and feature extraction. *J Big Data* 2024;11(1):33.